

7.15 block 07b split receipt — 2026-09-26

Decision and provenance

The owner authorized three stacked replacement review units after all three Copilot rounds on firmware PR #863 returned findings. PR #863 remains open as the historical complete candidate. Its final code head is 729ae7cfacf581ce534a98f6f5e0cc39efee3838, based on 8c654bcd8e1c8936a86aaa625a4ab78900a250e9. The reviews were 5323189342 (four findings), 5323944293 (two) and 5324788734 (one). All seven threads received source dispositions and are resolved. No fourth review is requested on the unsplit PR.

The #863 audit report and PDF were frozen before this authorization. Their "split pending" statement is a historical status; this receipt records the subsequent owner decision and replacement stack.

Replacement drafts: firmware [#865](https://github.com/BitHighlander/keepkey-firmware/pull/865), [#866](https://github.com/BitHighlander/keepkey-firmware/pull/866) and [#867](https://github.com/BitHighlander/keepkey-firmware/pull/867), with companion Python [#114](https://github.com/BitHighlander/python-keepkey/pull/114), [#115](https://github.com/BitHighlander/python-keepkey/pull/115) and [#116](https://github.com/BitHighlander/python-keepkey/pull/116).

Unit	Boundary	Recorded stack reference	Python companion
A-C formatters and interpolated intent	7a base	5c0090e038769ed36d627d4dca09cbfcb1df91d3	f813519c7c1163783b0e89b5cbd70536831f5e49
D groups, one-array iteration, optional visibility	A-C	83330002baf713173d65f6c8a2faf902716a6649	9ded843
E embedded calls and assembled audit	D	b611ea3a2 (runtime-control predecessor)	56e3585

The E row names its pre-receipt integration commit; its final report-bearing head is recorded in firmware PR #867. D includes the later wire-test expectation corrections and two fixes from its first split Copilot review, plus the updated official-registry signability assertion.

The E firmware and Python companion inherit D's later formatter array-binding and capture-initialization fixes. They therefore differ from historical #863 and companion #228 in those reviewed safety fixes. Before the D fixes, both source equivalences were checked with `git diff --exit-code`. Each unit adds only the source it compiles or dispatches; D and E source is absent from the A-C unit, and E source is absent from D.

File placement

Each cell identifies the adjacent branch diff that owns that file's edits. For a shared file, edits introduced in A-C, D and E are distinct changes; later edits to an earlier line belong to the later diff. The three adjacent diffs account for the entire original #863 tracked change set. The Python gitlink identifies three companion branches with the same boundaries.

Tracked path	A-C	D	E
deps/python-keepkey	✓	✓	✓
docs/release/audit-units/715-07b-formatters-audit-20260925.md			✓
docs/release/audit-units/715-07b-formatters-audit-20260925.pdf			✓
docs/security/HANDOFF-ERC7730-715-FORMATTERS.md	✓	✓	✓
docs/security/HANDOFF-ERC7730-PHASE-E.md			✓
include/keepkey/firmware/eip712_stream.h	✓		
include/keepkey/firmware/erc7730_abi_stream.h	✓		✓
include/keepkey/firmware/erc7730_capabilities.h	✓	✓	✓
include/keepkey/firmware/erc7730_catalog.h	✓	✓	✓
include/keepkey/firmware/erc7730_field.h	✓		✓
include/keepkey/firmware/erc7730_format.h	✓		

include/keepkey/firmware/erc7730_program.h	✓		
include/keepkey/firmware/erc7730_workflow.h	✓	✓	✓
include/keepkey/firmware/ethereum.h	✓		
lib/firmware/CMakeLists.txt	✓		
lib/firmware/eip712_stream.c	✓		
lib/firmware/erc7730_abi_stream.c	✓	✓	✓
lib/firmware/erc7730_capabilities.c	✓	✓	✓
lib/firmware/erc7730_catalog.c	✓	✓	✓
lib/firmware/erc7730_field.c	✓		✓
lib/firmware/erc7730_format.c	✓		
lib/firmware/erc7730_program.c	✓		
lib/firmware/erc7730_workflow.c	✓	✓	✓
lib/firmware/ethereum.c	✓		
lib/firmware/fsm.c			✓
lib/firmware/fsm_msg_ethereum.h	✓	✓	✓
scripts/emulator/test_stack07_regressions.py	✓	✓	✓
unittests/firmware/CMakeLists.txt	✓		
unittests/firmware/erc7730_abi_stream.cpp	✓		
unittests/firmware/erc7730_catalog.cpp	✓	✓	✓
unittests/firmware/erc7730_field.cpp	✓		✓
unittests/firmware/erc7730_format.cpp	✓		
unittests/firmware/erc7730_program.cpp	✓		
unittests/firmware/erc7730_workflow.cpp			✓

Finding placement and disposition

Finding group	Unit	Disposition
D1 numeric signer constants; D3 signer text and decimals; D4 native Wanchain state; D5 token table precedence; D6 intent titles; D7 line splitting; D8 control bytes and oversized raw values; D12 root signer path; D16 signer marks and raw unit	A-C	Preload refusal or exact on-screen disclosure, with native and wire controls.
D10 parallel array pairing; D18 fixed indices into dynamic arrays	D	Separate formatter arrays are refused before review; a fixed index absent from a dynamic array fails during capture. Firmware and mirror controls cover both.
D2 embedded intent; D9 literal callee; D17 refused inner definition; inner @.value without amountPath; depth-two and call binding	E	Refuse at preload or show blind, with inner-call and binding controls.
D11 table, text and decimal limits	A-C	The first companion mirrors preload bounds.
D11 iteration path limits	D	The second companion mirrors one-array iteration.
D11 embedded-call limits	E	The third companion mirrors inner-call capabilities.
D14 field title	D	Iteration fields use a numbered device-owned title.
D15 pager continuation test handling	E	Runtime tests filter continuation pages without de-duplicating them.
Copilot round 1: scalar formatter repeated in an iteration	D	Value path must walk the active array; positive/negative firmware and mirror tests.

Copilot round 1: stale phase counts, E status and EX test count	E	Handoffs and release report corrected.
Copilot round 2: display frames overwrote path classes	D	Dedicated <code>display_frames</code> , with a class-preservation test.
Copilot round 2: replay string reader accepted NUL	A-C	Independent reader rejects NUL, with a negative control.
Copilot round 3: signer native alias hid the token address	A-C	Signer alias and address precede the amount; maximum and Wanchain boundaries tested.
D split Copilot round 1: auxiliary formatter path escapes the active array	D, inherited by E	Reject mixed array or scalar formatter arguments inside iteration; firmware and Python three-case controls.
D split Copilot round 1: failed capture leaves count uninitialized	D, inherited by E	Initialize count before conditional capture read; full native suite and hosted CI.

The original 102-agent audit recorded 42 findings only as clusters D1–D19. The raw per-finding manifest was not retained. D13 and D19 were not named in the surviving handoff; this receipt cannot reconstruct or assign individual raw findings from those clusters. The 14 later re-audit defects are recorded in the #863 report as behaviors and controls, not as a raw manifest. These historical evidence gaps remain explicit; no finding is marked clean merely because its original row is unavailable. The seven Copilot comments are individually assigned above and resolved on #863.

Validation ledger

Unit	Local native evidence	Host/device compiler evidence	Other gates
A-C	firmware-unit 649/649 in a clean run directory	Focused compiler/validator controls pass	Exact-head CI 36270081889 passed; split Copilot round 1 had no inline findings.
D	firmware-unit 652/652 after review fixes	Compiler/validator 14 pass, 3 local skips; full registry 1,450 checked, 1,273 signable	CI 36283037667 passed all jobs. Split Copilot round 2 found a stale handoff count, fixed at <code>f4d3e63ae</code> ; exact-head CI at the latest D head and round 3 remain.
E	firmware-unit 656/656 on integrated D code	Compiler/catalog 24 pass, 3 local skips; full registry 1,450 checked, 1,297 signable; corrected emulator cases 2/2 pass	CI 36283243535 exposed two runtime tests for now-refused mixed-array definitions. The companion now tests a same-array embedded tuple and separate-array refusal; new exact-head CI required.

The canonical 7b pre-push script fails on inherited duplicate `nanopb` options, an absent report-gate test in the 7a base, and its own absent `.cppcheck-build` path. A corrected `cppcheck` invocation found no issues. The owner authorized the #863 push with this exception recorded. The split branches retain that same inherited preflight condition. Runs on all three split worktrees passed submodule cleanliness, unit-test registration and clang-format 20, then failed only those three inherited checks. Corrected `cppcheck` also completed without findings on A-C and D; E's integrated head needs its own hosted static result. Hosted CI and review results for replacement PRs belong to their exact heads and will be added to the PR descriptions.